

# LocalLens - A Crowd-Sourced Mobile Platform for Civic Issue Resolution in Urban Indian Communities

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**Abstract**—Urban infrastructure issues in urban areas often go unaddressed due to the lack of effective reporting channels between citizens and responsible authorities. This paper proposes the design of a mobile civic issue reporting platform that allows citizens to document local issues through in-app media capture, link reports to geopolitical jurisdictions, and share them publicly to pressure for social and political accountability. The proposed system supports anonymous posting, real-time messaging, a gamified contribution model, and dedicated public profiles for elected representatives. The application is currently under development. This paper presents the system concept, proposed architecture, module design, and planned implementation approach, along with considerations on security, privacy, and future scope.

**Index Terms** - Civic Engagement, Geopolitical Tagging, Anonymous Reporting, Mobile Application, Social Accountability, Gamification, Issue Reporting, Offline First, Real Time Communication, Representative Profile

## I. INTRODUCTION

Deterioration of civic infrastructure in densely populated areas and the lack of transparent channels for public-government communication remain persistent challenges. Residents often lack a structured system to report local issues, such as poor road conditions, illegal garbage dumping, or water supply problems, and receive clear information from responsible officials [1]. Outdated complaint portals fail to create the political accountability necessary for prompt resolution [2].

The rapid proliferation of smartphones has made mobile-first civic platforms a viable approach to participatory governance [3]. Platforms that leverage Geo-tagged media and public accountability records have demonstrated potential to improve municipal responsiveness [4]. However, a gap remains between

General purpose social media and purpose built civic tools: the former lack structured jurisdiction mapping, representative accountability features, and data integrity controls that a civic context demands.

This paper proposes a modular mobile civic issue reporting platform to address these shortcomings. The system allows citizens to capture and share Geo-tagged evidence of civic issues, assign them to the correct administrative jurisdiction, and interact with the public profiles of elected representatives. The platform includes anonymous posting, a gamified contribution model, real time messaging, and offline first data handling. The application is currently in development and has not been deployed.

## II. LITERATURE SURVEY

**Civic Engagement Platforms:** Digital civic platforms have been studied to bridge the gap between citizens and local governments. Offenhuber found that visibility of the problem and structured routing to responsible institutions are the most important factors for resolution outcomes [1]. Feedback loops that provide visible status updates to citizens are essential for continued engagement [5].

**Geo-tagged issue reporting:** Geo-tagging in civic platforms has been explored in the context of 311 service systems and the Open Government Initiative [6]. Geo-tagged reporting enables automated routing of complaints and supports specific visibility of problem density for evidence based resource allocation [4]. Linking a report to a specific ward or constituency, rather than just a single coordinate, provides a more targeted accountability system.

**Anonymous Reporting:** Anonymous participation reduces reporting barriers and also raises concerns about data quality [7]. Privacy preserving designs that separate user identities from report content while retaining location metadata for routing represent a recommended approach [8].

**Media Integrity:** Requiring media to be captured only within the application ensures that embedded GPS coordinates, timestamps, and device information reflect the actual conditions at the time of documentation [10], reducing the risk of fabricated or contextually misleading submissions [9].

**Gamification:** Point based rewards, leaderboards, and badge mechanisms improve sustained participation when designed to reward constructive behavior [11] [12]. The proposed platform adopts these principles in its coin and impact score systems.

**Real time communication:** WebSocket based persistent connections enable low latency message delivery in mobile applications [13]. Offline message queuing, in which messages are stored locally and transmitted upon reconnection, is a standard pattern in offline first designs [14]. Representative accountability: Making solution statistics and citizen ratings publicly accessible creates quantifiable social accountability that complements formal oversight mechanisms [16] [15].

### III. SYSTEM ARCHITECTURE

The proposed platform follows a three-tier feature architecture, where each functional module is divided into a data layer for API communication and local persistence, a domain layer for business logic and repository interfaces, and a presentation layer for user interface state and presentation [17]. This separation of concerns allows individual modules to be developed and extended independently.

The application is built using Flutter, targeting Android as the primary environment while maintaining cross-platform compatibility. The project adopts a feature first folder structure, in which each feature has its own data, domain, and presentation sublayers. Cross-cutting concerns such as theming, routing, and error handling are contained in shared core modules. A shared widget library enforces visual consistency across the application.

Given that a portion of the target user base operates in low-connectivity environments, the platform is designed with offline first considerations. Post drafts, outgoing messages, and pending media uploads are stored in local persistent storage and synchronized with the backend upon reconnection [14]. The messaging subsystem uses a persistent real time connection with an offline message queue that sends stored messages in order when connectivity is restored.

The backend supports a number of microservices for user authentication, post management, media storage, notifications, analytics, and messaging. A dedicated analytics module, described in the MVP, keeps track of each user's activity, including the number of posts submitted and the number of issues resolved.

### IV. SYSTEM DESIGN AND MODULES

**Authentication:** The module supports phone number and email based sign in with one time password verification. Anonymous browsing with restricted features is available without registration. Multiple accounts can be stored locally and switched from the settings interface.

**Home Feed:** The feed presents an infinite scrolling stream of Geo-relevant posts. Each entry displays a jurisdiction location tag, an issue status indicator (active, resolved, under review, or urgent), related media, the author's display name or "anonymous" label, and interaction controls including upvote, downvote, comment, share, and flag. A continuous bottom navigation bar provides access to home, search, post creation, notifications, and messages.

**Post Timeline:** Tapping on a post's media opens a full screen chronological timeline of the issue. Each entry contains a timestamp, status or update description, and optionally a new media attachment, providing a complete audit trail from initial report to resolution.

**Create Post and Draft Composer:** Opens the device camera directly in the post creation application. GPS coordinates, timestamps, and device metadata are automatically embedded at the capture point. Gallery import is not provided as this constraint is fundamental to metadata integrity [10]. Draft Composer allows the user to add a title, caption, searchable tags, and text overlays, choose public or anonymous visibility, group multiple images in a carousel, and tag users or representative profiles before publishing.

**Search, Explore, and Map View:** The search module supports keyword, location, and username queries with status and proximity filter chips. Results are displayed as thumbnail cards. The map view toggle plots all results as clustered geographic pins. Tapping on a pin opens a sheet preview at the bottom with a link to the entire post timeline. The map supports filtering by issue type, status, and date range, and allows for searching areas independent of the user's own location.

**Notifications:** Alerts are grouped by category, including upvotes, comments, follows, issue status updates, mentions, and system messages. Individual and bulk read actions are provided. Each notification deeplinks to the relevant post, comment thread, or user profile. **Messaging and Chat:** Conversations are listed in the inbox with avatar, display name, last message preview, unread badge, timestamp, and online status indicator. Individual and group chats support real time exchanges with delivery statuses of sent, delivered, and read. Media shared in chats must be captured in the application, consistent with the platform's media integrity policy.

**User Profile and Profile Management:** The public profile displays the user's photograph, username or anonymous label, biography, and unique identifier. The profile management screen allows editing of personal details,

post management, and per post or global anonymous mode configuration.

**Representative Profile:** Verified profiles for elected representatives display a jurisdiction descriptor indicating the covered ward, district, or constituency. The profile presents issues raised in that jurisdiction, the number resolved, resolution rate, and average resolution time. Citizens may submit a public rating contributing to an aggregated score visible on the profile.

**Gamification:** Users earn coins for submitting issue reports, receiving upvotes, confirming that an issue is resolved, and properly tagging the representative responsible. Area level leaderboards rank users based on their civic contributions in their ward or constituency. An impact score, calculated as a composite of community reach and resolution rate, is displayed on the public profile, along with badges for milestones such as first issue submitted, tenth issue submitted, resolution champion, and top area contributor. The gamification module is structured as a separate component to allow expansion without modifying other modules.

**Analytics:** A dedicated analytics table stores each user's activity data, including post counts and resolved issue counts. This data supports impact score calculations, representative profile statistics population, and future analytics capabilities.

**Settings:** The Settings interface is divided into subpages for account management, privacy configuration, notification preferences, application permissions including current status display, session activity, appearance, help and support, and information. Account management supports adding and switching between multiple locally stored accounts.

## V. IMPLEMENTATION PLAN

The MVP specification says that the development should happen in four stages. The first phase builds the basic infrastructure, which includes authentication, the home feed, the in app camera, the draft composer, and the flow for submitting Geo-tagged posts with embedded metadata. These parts make up the minimum set of parts the platform needs to do its main civic reporting tasks.

The second phase adds the social interaction layer, which includes comments, upvotes and downvotes, a follower system, notifications, search with a map view, and representative profiles with search and jurisdiction tagging.

The third phase has real time messaging with offline queuing, group chat, media capture in chat, and a full gamification module with coins, leaderboards, badges, and impact score calculations.

The fourth phase includes the analytics module, a full settings interface, multi account support, content validation, media compression for low bandwidth environments, and performance profiling for deployment readiness.

## VI. SECURITY AND PRIVACY CONSIDERATIONS

**Anonymous Posting:** When anonymous mode is active, the author field is replaced with the label "Anonymous" and

no personal identifiers appear in public-facing post data [8]. Location metadata is retained for jurisdiction routing but is presented at the administrative area level rather than as a precise coordinate in the public display. Identity decoupling is enforced at the data layer to prevent re-identification through the presentation layer.

**Media Metadata Integrity:** GPS coordinates, timestamps, and device information are embedded in the app at the point of capture and cannot be changed before submission [10]. This prevents misrepresentation of the location or time of a reported incident [9].

**Authentication and Session Management:** One time password verification via phone or email mitigates the risks associated with static credentials. The Session Activity subpage displays login history and active sessions, allowing users to identify and terminate unauthorized access.

**Data Redaction and Consent:** Personal data is not disclosed without explicit user consent. The privacy settings interface provides granular control over data sharing. Anonymous browsing limits data collection to only what is technically necessary to provide the content feed.

**Content Validation:** Server side validation is used to detect errors in metadata and flag submissions that exhibit different patterns, thereby supporting a trustworthy civic information environment.

## VII. FUTURE SCOPE

Integration with formal government grievance redressal systems will allow verified reports to be sent directly to the official grievance portal, integrating the platform's social accountability mechanism into the legal resolution workflow through a standardized data exchange interface.

Machine learning based problem classification can automatically classify submitted reports by problem type based on media content and related text, reducing citizen effort in submissions and improving consistency of jurisdictional routing.

Predictive civic analytics based on accumulated historical data can reveal patterns in problem frequency by area, time period, and problem type, helping municipal planners anticipate recurring problems.

Improved jurisdictional mapping will extend the geopolitical tagging layer to cover finer administrative subdivisions and accommodate different structures within states, improving the accuracy of assignment of responsibility.

Interface localization and multilingual accessibility improvements, including common language support for captions and search queries, will expand the platform's reach to citizens whose primary language is not English.

## VIII. CONCLUSION

This paper demonstrates the design of a modular mobile civic issue reporting platform that bridges the gap between citizens' experiences of local infrastructure problems and the accountability systems that exist to address them. The proposed system integrates geo-political jurisdiction tagging,

in-application media capture with automatic metadata embedding, anonymous posting, representative public profiles, a gamified participation model, and real-time messaging within a single offline-capable application.

The platform is currently in development. The three-tier modular architecture separates data access, business logic, and presentation concerns within each feature module, allowing for independent development and future expansion. The security and privacy design prioritizes user identity protection and media integrity, both of which are essential in citizen reporting, where disclosure could pose a social or political threat. Empirical evaluation following deployment will provide further insight into the solution's results to guide adoption methods and subsequent design iterations.

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